

Sets & logic

(unordered)

A set is a collection of objects.

$\{0, 1, 5\}$

$\{5, 0, 1\}$

$S = \{(0, 0), \text{Car}, \text{orange}, \text{Chocolate}\}$

$\mathbb{N}$: natural numbers

$\mathbb{R}$: real numbers

$\mathbb{Q}$: rational numbers

$\mathbb{C}$: complex numbers

We say "x belongs to A" and write $x \in A$.

$\{1, 2, 3, 4, \dots\}$
 $\rightarrow \text{keep}$

$x \leq 3$

$$S = \{x \in \mathbb{R} \mid x^2 - 2x + 1 \geq 0\} \quad S = \{x \in \mathbb{R} \mid x \leq 3\}$$

$$P = \{p \in \mathbb{N} \mid p \text{ is prime}\}$$

PREDICATE

$$\begin{array}{c} 3 \leq 3 \\ \swarrow \quad \searrow \\ 3 < 3 \quad \text{or} \quad 3 = 3 \end{array}$$

Definitions:

① We say $A \subseteq B$ if every member of A is also in B.

Logical statement:

$$\left[x \leq 3 \text{ \& } (x^2 - 2x + 2 \leq 0) \right] \text{ OR } x > 5$$

$$\begin{array}{l} x \leq 3 \Rightarrow x \leq 5 \\ \{x \mid x \leq 3\} \subseteq \{x \mid x \leq 5\} \end{array}$$

$\emptyset \subseteq A$ for every A
 $\emptyset$: empty set

We say that

$x \in A \Rightarrow x \in A$

Statement $A \Rightarrow$ Statement B
 if $\{x \mid A \text{ holds}\} \subseteq \{x \mid B \text{ holds}\}$

$x \in \emptyset \Rightarrow x \in \Pi$
 $\uparrow$
 Vacuous reasoning

② we say that $A=B$ if $A \subseteq B$ and $B \subseteq A$.

$\emptyset \subseteq \emptyset'$
 $\emptyset' \subseteq \emptyset$
 $\Rightarrow \emptyset = \emptyset'$

$$A = \{x \mid x \in A\}$$

Complements:

$$A^c = \{x \mid x \notin A\}$$

"not in"
 "does not belong to".

Universe / context

$$x \in A \Rightarrow x \in B$$

Property 1: 1) $A \subseteq B$

These statements are equivalent -

2) $B^c \subseteq A^c$

$A \subseteq B$ if & only if $B^c \subseteq A^c$
 iff

Proof: Statement 1 $\Rightarrow$ Statement 2

Let $x \in B^c$. Then $x \notin B \Rightarrow x \notin A$
 $\Rightarrow x \in A^c$

$B^c \subseteq A^c$

Statement 2 $\Rightarrow$ Statement 1

(Exercise)

Property 2: If Statement 1 $\Rightarrow$ Statement 2

then

$\text{NOT}(\text{statement } 2) \Rightarrow \text{NOT}(\text{statement } 1)$

Contrapositive.

Property 3: $A \subseteq B, B \subseteq C$ then $A \subseteq C$

$$x \in A \Rightarrow x \in B \Rightarrow x \in C$$

Defn: (Tautology): A statement that is always true regardless of the subject.

$$x \leq 3 \text{ or } x > 3$$

(Contradiction): A statement that is always false regardless of the subject

$$x \leq 3 \text{ AND } x > 3$$

If $A \Rightarrow$ contradiction then A is a contradiction.

$$\underbrace{x^2 - 2x + 1}_{\uparrow} < 0 \Rightarrow (x-1)^2 < 0$$

$\Rightarrow$ a perfect square is negative

$\uparrow$

$$x \leq 3 \Rightarrow x \leq \infty$$

$$x \leq 3 \Rightarrow \underbrace{x \in \mathbb{R}}$$

Proof by contradiction: Prove a statement S is true.

Technique: $\text{NOT}(S) \Rightarrow \text{contradiction}$.

Example 1: the number of primes is infinite } S

Suppose the number of primes were finite. } $\text{NOT}(S)$

Let $p_1, p_2, \dots, p_n$ be the primes.

$$\text{Let } p = 1 + p_1 p_2 p_3 \dots p_n$$

then, p can only have 1 & p as divisors.

$\Rightarrow p$ is prime

a contradiction.

Example 2: the largest integer is 1 } S

Suppose the largest integer is $n > 1$. Then $n^2 = n \times n$

$$\text{then } n^2 > n$$

a contradiction.

Hence, the largest integer is 1.

Recap: ① A mathematical statement is a "predicate" $P(x)$ which returns true or false for various "subjects" x .

② we say $P(x) \Rightarrow Q(x)$ If then
 "Implies" $x > 0 \Rightarrow x > -2$

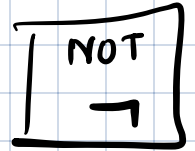
Let $S_P = \{x \mid P(x)\}$
 $S_Q = \{x \mid Q(x)\}$

$P(x) \Rightarrow Q(x)$
 $S_P \subseteq S_Q$

$\neg P(x)$

③ $A \subseteq B$ $B^c \subseteq A^c$
 Contraposition

$P(x) \Rightarrow Q(x)$
 $\text{NOT}(Q(x)) \Rightarrow \text{NOT}(P(x))$



④ Tautologies & Contradictions

always true
 regardless of
 subject-
 in the given universe/context

statement
 always false regardless of
 subject.

⑤ $P(x) \Rightarrow$ A contradiction then $P(x)$ is a contradiction.
 used often in "proof by contradiction".

Composite statements

AND

OR

$P(x)$ AND $Q(x)$
 $\&$

$P(x)$ OR $Q(x)$

Truth tables

T

$P(x)$	$Q(x)$	$P(x) \& Q(x)$
True	True	True
True	False	False
False	True	False
False	False	False

"for all" $\forall$

$$\underline{P(x)} : x^2 \geq 0 \quad \forall x \in \mathbb{R}$$

$$\text{NOT}(P(x)) : \exists x \in \mathbb{R} : x^2 < 0$$

↑ ↑
there exists such that ($:$, $|$, $\exists$)

$\&$ $\longleftrightarrow$ intersection

$$A \cap B$$

$$x \in A \ \& \ x \in B \Rightarrow x \in A \cap B.$$

$P(x)$ OR $Q(x)$	$P(x)$	$Q(x)$	$P(x) \text{ OR } Q(x)$
	true	true	true
	false	true	true
	true	false	true
	false	false	false

OR $\longleftrightarrow$ Unions.

$$A \cup B$$

$$x \in A \text{ or } x \in B \Rightarrow x \in A \cup B.$$

De Morgan's law :

$$(A \cup B)^c = B^c \cap A^c$$

$$\text{NOT}(P \text{ OR } Q) \iff (\text{NOT } P) \text{ AND } (\text{NOT } Q)$$

equivalent
to

Proof outline : $(A \cup B)^c = B^c \cap A^c$

Part 1: $(A \cup B)^c \subseteq B^c \cap A^c$

Part 2: $B^c \cap A^c \subseteq (A \cup B)^c$

Let $x \in (A \cup B)^c$

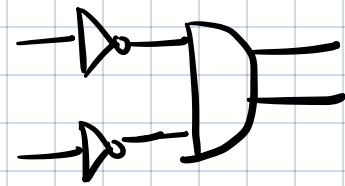
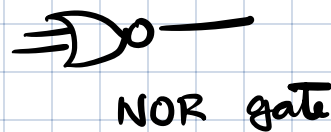
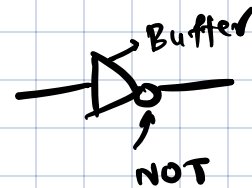
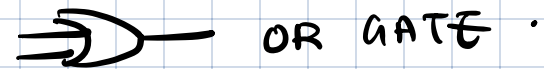
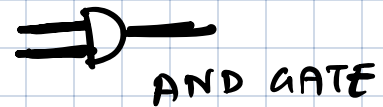
then $x \notin A \cup B$

then $x \notin A, x \notin B$

then $x \in A^c, x \in B^c$

then $x \in A^c \cap B^c$

Exercise !



Multiple compositions

$$(P \& Q) \& R = P \& (Q \& R)$$

Associativity

real nos add,
mult
matrix mult

$$A: 1 \times 10^6$$

$$B: 10^6 \times 1$$

$$C: 1 \times 1$$

① $A(B \& C)$

② $(A \& B) \& C$ (cheaper)

Exercise:

How many mult & add are needed to compute ① vs ②?

OR, AND, $\cup$, $\cap$

satisfy associativity

$$A \cap (B \cap C) = (A \cap B) \cap C, A \cup (B \cup C) = (A \cup B) \cup C$$

Commutativity:

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

Distributivity:

$$\textcircled{1} A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$a(b+c) = ab+ac$$

$$a+bc$$

$$\neq (a+b)(a+c)$$

Exercise : Prove $\textcircled{1}$.

More general De Morgan's law: Any set-theoretic statement involving ONLY unions & intersections continues to hold if the $\cup$ & $\cap$ symbols are swapped.

Example 1: $S: \underbrace{x^2 + 2xy + y^2}_{\geq 4xy}$

$$\forall x, y \in \mathbb{R}$$

$$\left. \begin{aligned} x^2 - 2xy + y^2 &\geq 0 \\ (x-y)^2 &\geq 0 \\ \text{which is true,} \end{aligned} \right\} \swarrow$$

$$S \Rightarrow \text{tautology}$$

$$\left. \begin{aligned} (x-y)^2 &\geq 0 \\ \text{then } x^2 - 2xy + y^2 &\geq 0 \\ \text{then } x^2 + 2xy + y^2 &\geq 4xy \end{aligned} \right\}$$

$$\text{Tautology} \Rightarrow S$$

$A: 2 \times 2 \text{ matrix}$

Example 2: If $A\underline{x} = \underline{0} \quad \forall \quad \underline{x} \in \mathbb{R}^2$, then $A = \underline{0}$.

$$\text{If } \underline{x} = \underline{0}, \text{ then } A\underline{x} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \underline{0}$$

Approach 1: - prove by contradiction.

$\exists$ a non-zero matrix A such that $A\underline{x} = \underline{0} \quad \forall \quad \underline{x}$

Suppose the top-left entry in A is non-zero.

$$\begin{pmatrix} a \neq 0 & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} \neq \underline{0}$$

Suppose the top-right entry in A is non-zero

$$\text{then } \begin{pmatrix} a & b \neq 0 \\ c & d \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} \neq \underline{0}.$$

Similarly for bottom left & bottom right.

Approach 2: direct proof

$$A\underline{x} = \underline{0} \quad \forall \quad \underline{x} \in \mathbb{R}^2.$$

$$\text{then } A\underline{x}_1 = \underline{0} \quad A\underline{x}_2 = \underline{0} \quad \text{for } \underline{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$$\text{then } \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\downarrow} = \underline{0}, \quad \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{\downarrow} = \underline{0} \quad \text{for } \underline{x}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

then $a=0, c=0$

$b=0, d=0$

then $A=0$.

Example 3: If n is odd, so is n^2 .

$$\underbrace{n \text{ odd}}_P \Rightarrow \underbrace{n^2 \text{ odd}}_Q$$

Let n be odd.

then $n = 2k+1$ for some integer k .

$$\begin{aligned} \text{then } n^2 &= (2k+1)^2 = 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \end{aligned}$$

then $n^2 = 2k' + 1$ for some integer k'

thus n^2 is odd.

Example 4: If n^2 is even, so is n .

$$\underbrace{\text{not}(Q)}_{n^2 \text{ is even}} \Rightarrow \text{not}(P)_{n \text{ is even}}$$

(Contraposition
of 3)

$$n^2 \text{ is even} \Rightarrow n \text{ is even.}$$

Example 5: For integers a, b, c ;
If $\underbrace{a^2 + b^2 = c^2}_P$ then at least one of a, b, c is even.
 $\underbrace{\hspace{10em}}_Q$

Approach 1:

Suppose $a^2 + b^2 = c^2$

and $\{a \text{ \& } b \text{ are odd}\} \rightarrow a^2 \text{ \& } b^2 \text{ are odd} \Rightarrow c^2 \text{ is even} \Rightarrow c \text{ is even}$

$\left. \begin{array}{l} a \& b \text{ are even/odd} \\ a \& b \text{ are even/even} \\ a \& b \text{ are odd/even.} \end{array} \right\} \Rightarrow Q \text{ is true.}$
 The statement Q is true.

Contrapositive:

a, b, c are odd $\Rightarrow a^2 + b^2 \neq c^2$.

Suppose a, b, c are odd $\& a^2 + b^2 = c^2$

then $a^2 + b^2 = c^2$ is even

then c is even.

leading to a Contradiction

Example 6: $\sqrt{2}$ is irrational.

Contradiction: Suppose $\sqrt{2}$ were rational.

then $\sqrt{2} = \frac{m}{n}$, integers m, n .

where m, n have no common factors.

then $2 = \frac{m^2}{n^2} \Rightarrow m^2 = 2n^2$

then m is even

then m is even.

then $m = 2k$ for some k

then $m^2 = 4k^2 = 2n^2$

then $n^2 = 2k^2$

then n^2 is even

then n is even.

thus m, n are both even, contradicting the
defn of m, n .

Example 7:

If $y \leq x + \epsilon$ $\underbrace{\forall \epsilon > 0}_{\text{For all}}$ then $y \leq x$.

Proof: $y \leq x + \epsilon$ $\forall \underbrace{\epsilon > 0}$

then $y \leq x + 0$

Example 8: If $\underbrace{y \leq x + \epsilon \quad \forall \epsilon > 0}_p$ then $\underbrace{y \leq x}$.

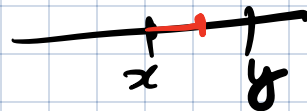
Proof:

Contrapositive

$$\text{not}(Q) \Rightarrow \text{not}(P)$$

Assume $\text{not}(Q)$: i.e. $y > x$.

$$\text{Let } \epsilon = \frac{y-x}{2}$$



then

$$x + \epsilon$$

$$= \frac{y-x}{2} + x$$

$$= \frac{y+x}{2} < \frac{y+y}{2}$$

$$= y$$

not(P):
 $\exists \epsilon > 0$ such that
 $y > x + \epsilon$
 $\epsilon = \frac{y-x}{2}$

If $y > x$, then we obtain an ϵ such that

$$y > x + \epsilon$$

$$\text{not}(Q) \Rightarrow \text{not}(P).$$

Example 8: For any real number x ,

either $\sqrt{3} + x$ or $\sqrt{3} - x$ is irrational.

$$\text{Let } \sqrt{3} + x \text{ is rational. Then } \sqrt{3} = \text{rational} - x$$

Suppose both $\sqrt{3} + x$ and $\sqrt{3} - x$ are rational

then $\frac{\sqrt{3} + x + \sqrt{3} - x}{2}$ is rational

then $\sqrt{3}$ is rational,

which is a contradiction.

Example 9: (Optional)

there is no smallest rational number above $\sqrt{2}$.

Archimedes
principle:
For any real
 x , $\exists$ integer n
such that $n > x$.